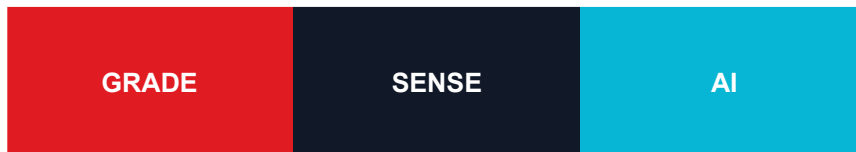


GradeSense API Documentation

Complete REST and WebSocket reference with generated request/response examples



Explainable predictions, constraint-aware recommendations, and real-time operations for safer, faster paper grade transitions.

Prepared from the validated GradeSense v0.2.0 source repository and live application.

Generated from the live OpenAPI 3 schema. HTTP endpoints documented: 47.

API Conventions

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- Base URL: deployment-specific; local reference `http://localhost:8000`.
- JSON fields use `snake_case`; timestamps are ISO 8601 UTC; IDs are UUID strings.
- Validation failures return 422, missing resources 404, and invalid lifecycle conflicts 409.
- Every HTTP response carries X-Request-ID and security headers.
- Interactive documentation is available at `/docs` and `/redoc`.

GET /

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API root

Response example - HTTP 200

```
{
  "name": "string",
  "message": "string",
  "documentation_url": "string"
}
```

GET /health

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Service health

Response example - HTTP 200

```
{
  "status": "string",
  "service": "string",
  "version": "string",
  "environment": "string"
}
```

GET /version

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Application version

Response example - HTTP 200

```
{
  "name": "string",
  "version": "string"
}
```

POST /predict

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Predict transition quality

Request example

```
{
  "current_grade": "string",
  "target_grade": "string",
  "machine_speed": 0.5,
  "steam_pressure": 0.5,
  "dryer_temperature": 0.5,
  "moisture": 0.5,
  "basis_weight": 0.5,
  "caliper": 0.5,
  "pulp_consistency": 0.5,
  "stock_flow": 0.5,
  "refining_energy": 0.5,
  "headbox_pressure": 0.5
}
```

Response example - HTTP 200

```
{
  "prediction_id": "00000000-0000-4000-8000-000000000001",
  "quality_score": 0.5,
  "off_spec_probability": 0.5,
  "expected_stabilization_time": 0.5,
  "model_version": "string",
  "explanation": {
    "summary": "string",
    "top_contributing_features": [
      "...",
    ],
    "feature_importance": {}
  },
  "created_at": "2026-07-26T12:00:00Z"
}
```

POST /recommend

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Generate model-evaluated corrective actions

Request example

```
{
  "current_grade": "string",
  "target_grade": "string",
  "machine_speed": 0.5,
  "steam_pressure": 0.5,
  "dryer_temperature": 0.5,
  "moisture": 0.5,
  "basis_weight": 0.5,
  "caliper": 0.5,
  "pulp_consistency": 0.5,
  "stock_flow": 0.5,
  "refining_energy": 0.5,
  "headbox_pressure": 0.5
}
```

Response example - HTTP 200

```
{
  "recommendation_id": "00000000-0000-4000-8000-000000000001",
  "prediction": {
    "prediction_id": "00000000-0000-4000-8000-000000000001",
    "quality_score": 0.5,
    "off_spec_probability": 0.5,
    "expected_stabilization_time": 0.5,
    "model_version": "string",
    "explanation": "...",
    "created_at": "2026-07-26T12:00:00Z"
  },
  "recommendations": [
    {
      "text": "...",
      "confidence": "...",
      "expected_improvement": "...",
      "affected_variables": "...",
      "inference_sources": "..."
    }
  ],
  "created_at": "2026-07-26T12:00:00Z"
}
```

GET /correlations

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Discover process correlations

Parameter	Location	Required	Description
limit	query	no	integer

Response example - HTTP 200

```
{
  "record_count": 1,
  "correlation_matrix": {},
  "strongest_positive_correlations": [
    {
      "first_variable": "...",
      "second_variable": "...",
      "correlation": "..."
    }
  ],
  "strongest_negative_correlations": [
    {
      "first_variable": "...",
      "second_variable": "...",
      "correlation": "..."
    }
  ]
}
```

GET /dataset/statistics

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Describe the generated industrial dataset

Response example - HTTP 200

```
{
  "record_count": 1,
  "generated_at": "2026-07-26T12:00:00Z",
  "columns": [
    "string"
  ],
  "missing_values": {},
  "numeric_summary": {},
  "grade_distribution": {}
}
```

POST /dataset/regenerate

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Regenerate the dataset and retrain the model

Request example

```
{
  "records": 20000,
  "seed": 42
}
```

Response example - HTTP 201

```
{
  "records": 1,
  "dataset_path": "string",
  "model_version": "string",
  "training_metrics": {}
}
```

Describe the active trained model

Response example - HTTP 200

```
{
  "model_type": "string",
  "model_version": "string",
  "trained_at": "2026-07-26T12:00:00Z",
  "training_records": 1,
  "feature_count": 1,
  "target_metrics": {},
  "dataset_checksum": "string",
  "supported_outputs": [
    "string"
  ],
  "artifact_path": "string"
}
```

List persisted predictions

Parameter	Location	Required	Description
page	query	no	integer
page_size	query	no	integer

Response example - HTTP 200

```
{
  "items": [
    {
      "prediction_id": "...",
      "model_version": "...",
      "input_data": "...",
      "quality_score": "...",
      "off_spec_probability": "...",
      "expected_stabilization_time": "...",
      "explanation": "...",
      "created_at": "..."
    }
  ],
  "pagination": {
    "page": 1,
    "page_size": 1,
    "total": 1,
    "total_pages": 1
  }
}
```

List persisted recommendations

Parameter	Location	Required	Description
page	query	no	integer
page_size	query	no	integer

Response example - HTTP 200

```
{
  "items": [
    {
      "recommendation_id": "...",
      "prediction_id": "...",
      "recommendations": "...",
      "created_at": "..."
    }
  ],
  "pagination": {
    "page": 1,
    "page_size": 1,
    "total": 1,
    "total_pages": 1
  }
}
```

Alerts

Parameter	Location	Required	Description
acknowledged	query	no	
limit	query	no	integer

Response example - HTTP 200

```
[
  {
    "id": "00000000-0000-4000-8000-000000000001",
    "severity": "info",
    "title": "string",
    "description": "string",
    "timestamp": "2026-07-26T12:00:00Z",
    "affected_variables": [
      "string"
    ],
    "suggested_action": "string",
    "acknowledged": true,
    "acknowledged_at": "2026-07-26T12:00:00Z",
    "prediction_id": "00000000-0000-4000-8000-000000000001"
  }
]
```

POST /alerts/{alert_id}/acknowledge

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Acknowledge Alert

Parameter	Location	Required	Description
alert_id	path	yes	string

Response example - HTTP 200

```
{
  "id": "00000000-0000-4000-8000-000000000001",
  "severity": "info",
  "title": "string",
  "description": "string",
  "timestamp": "2026-07-26T12:00:00Z",
  "affected_variables": [
    "string"
  ],
  "suggested_action": "string",
  "acknowledged": true,
  "acknowledged_at": "2026-07-26T12:00:00Z",
  "prediction_id": "00000000-0000-4000-8000-000000000001"
}
```

GET /feedback

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Feedback

Response example - HTTP 200

```
[
  {
    "prediction_id": "00000000-0000-4000-8000-000000000001",
    "outcome": "recommendation_accepted",
    "notes": "string",
    "id": "00000000-0000-4000-8000-000000000001",
    "created_at": "2026-07-26T12:00:00Z"
  }
]
```

POST /feedback

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Create Feedback

Request example

```
{
  "prediction_id": "00000000-0000-4000-8000-000000000001",
  "outcome": "recommendation_accepted",
  "notes": "string"
}
```

Response example - HTTP 201

```
{
  "prediction_id": "00000000-0000-4000-8000-000000000001",
  "outcome": "recommendation_accepted",
  "notes": "string",
  "id": "00000000-0000-4000-8000-000000000001",
  "created_at": "2026-07-26T12:00:00Z"
}
```

GET /stream/status

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Stream Status

Response example - HTTP 200

```
{
  "status": "starting",
  "session_id": "00000000-0000-4000-8000-000000000001",
  "started_at": "2026-07-26T12:00:00Z",
  "sample_count": 1,
  "connected_clients": 1,
  "latest_sample_at": "2026-07-26T12:00:00Z"
}
```

GET /metrics/rolling

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Rolling Metrics

Response example - HTTP 200

```
[
  {
    "window": "string",
    "average_quality": 0.5,
    "average_off_spec_probability": 0.5,
    "average_stabilization_time": 0.5,
    "recommendation_frequency": 1,
    "alert_frequency": 1,
    "prediction_count": 1
  }
]
```

GET /stream/statistics

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Rolling Metrics

Response example - HTTP 200

```
[
{
  "window": "string",
  "average_quality": 0.5,
  "average_off_spec_probability": 0.5,
  "average_stabilization_time": 0.5,
  "recommendation_frequency": 1,
  "alert_frequency": 1,
  "prediction_count": 1
}
```

GET /metrics/live

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Live Metrics

Response example - HTTP 200

```

{
  "sensor": {
    "current_grade": "...",
    "target_grade": "...",
    "machine_speed": "...",
    "steam_pressure": "...",
    "dryer_temperature": "...",
    "moisture": "...",
    "basis_weight": "...",
    "caliper": "...",
    "pulp_consistency": "...",
    "stock_flow": "...",
    "refining_energy": "...",
    "headbox_pressure": "..."
  },
  "prediction": {
    "prediction_id": "...",
    "quality_score": "...",
    "off_spec_probability": "...",
    "expected_stabilization_time": "...",
    "model_version": "...",
    "explanation": "...",
    "created_at": "..."
  },
  "recommendations": [
    {
      "text": "...",
      "confidence": "...",
      "expected_improvement": "...",
      "affected_variables": "...",
      "inference_sources": "..."
    }
  ],
  "alerts": [
    {
      "id": "...",
      "severity": "...",
      "title": "...",
      "description": "...",
      "timestamp": "...",
      "affected_variables": "...",
      "suggested_action": "...",
      "acknowledged": "...",
      "acknowledged_at": "...",
      "prediction_id": "..."
    }
  ],
  "drift": {
    "score": "...",
    "severity": "...",
    "drifting_variables": "...",
    "prediction_drift": "...",
    "recommended_action": "...",
    "calculated_at": "..."
  },
  "updated_at": "2026-07-26T12:00:00Z"
}

```

GET /drift

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Drift

Response example - HTTP 200

```
{
  "score": 0.5,
  "severity": "stable",
  "drifting_variables": {},
  "prediction_drift": 0.5,
  "recommended_action": "string",
  "calculated_at": "2026-07-26T12:00:00Z"
}
```

POST /forecast

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Create Forecast

Request example

```
{
  "transition_id": "string",
  "current_grade": "string",
  "target_grade": "string",
  "target_basis_weight": 0.5,
  "history": [
    {
      "timestamp": "...",
      "timestep": "...",
      "stock_flow": "...",
      "filler_flow": "...",
      "steam_pressure": "...",
      "machine_speed": "...",
      "dryer_temperature": "...",
      "moisture": "...",
      "ash": "...",
      "caliper": "...",
      "basis_weight": "...",
      "reel_tension": "..."
    }
  ],
  "forecast_horizon": 1
}
```

Response example - HTTP 200

```
{
  "forecast_id": "00000000-0000-4000-8000-000000000001",
  "transition_id": "string",
  "model_version": "string",
  "history_window": 1,
  "forecast_horizon": 1,
  "confidence": 0.5,
  "trajectory": [
    {
      "step": "...",
      "timestamp": "...",
      "basis_weight": "...",
      "lower_bound": "...",
      "upper_bound": "...",
      "deviation_pct": "...",
      "lower_spec_limit": "...",
      "upper_spec_limit": "..."
    }
  ],
  "specification": {
    "target_basis_weight": 0.5,
    "lower_spec_limit": 0.5,
    "upper_spec_limit": 0.5,
    "current_deviation_pct": 0.5,
    "maximum_predicted_deviation_pct": 0.5,
    "crossing_probability": 0.5,
    "predicted_crossing_step": "...",
    "predicted_crossing_time": "...",
    "remaining_safe_operating_seconds": "...",
    "predicted_stabilization_step": "..."
  },
  "top_influencing_variables": [
    null
  ],
  "explanation": "string",
  "created_at": "2026-07-26T12:00:00Z"
}
```

GET /forecast/history

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Forecast History

Parameter	Location	Required	Description
limit	query	no	integer

Response example - HTTP 200

```
{
  "items": [
    {
      "forecast_id": "...",
      "transition_id": "...",
      "model_version": "...",
      "history_window": "...",
      "forecast_horizon": "...",
      "confidence": "...",
      "trajectory": "...",
      "specification": "...",
      "top_influencing_variables": "...",
      "explanation": "...",
      "created_at": "..."
    }
  ],
  "total": 1
}
```

GET /forecast/{forecast_id}

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Get Forecast

Parameter	Location	Required	Description
forecast_id	path	yes	string

Response example - HTTP 200

```
{
  "forecast_id": "00000000-0000-4000-8000-000000000001",
  "transition_id": "string",
  "model_version": "string",
  "history_window": 1,
  "forecast_horizon": 1,
  "confidence": 0.5,
  "trajectory": [
    {
      "step": "...",
      "timestamp": "...",
      "basis_weight": "...",
      "lower_bound": "...",
      "upper_bound": "...",
      "deviation_pct": "...",
      "lower_spec_limit": "...",
      "upper_spec_limit": "..."
    }
  ],
  "specification": {
    "target_basis_weight": 0.5,
    "lower_spec_limit": 0.5,
    "upper_spec_limit": 0.5,
    "current_deviation_pct": 0.5,
    "maximum_predicted_deviation_pct": 0.5,
    "crossing_probability": 0.5,
    "predicted_crossing_step": "...",
    "predicted_crossing_time": "...",
    "remaining_safe_operating_seconds": "...",
    "predicted_stabilization_step": "..."
  },
  "top_influencing_variables": [
    [
      null
    ]
  ],
  "explanation": "string",
  "created_at": "2026-07-26T12:00:00Z"
}
```

POST /forecast/simulate

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Simulate Forecast

Request example

```
{
  "forecast_id": "00000000-0000-4000-8000-000000000001",
  "changes": [
    {
      "variable": "...",
      "value": "..."
    }
  ]
}
```

Response example - HTTP 201

```
{
  "simulation_id": "00000000-0000-4000-8000-000000000001",
  "forecast_id": "00000000-0000-4000-8000-000000000001",
  "recommendation_id": "00000000-0000-4000-8000-000000000001",
  "changes": [
    {
      "variable": "...",
      "value": "..."
    }
  ],
  "baseline_trajectory": [
    {
      "step": "...",
      "timestamp": "...",
      "basis_weight": "...",
      "lower_bound": "...",
      "upper_bound": "...",
      "deviation_pct": "...",
      "lower_spec_limit": "...",
      "upper_spec_limit": "..."
    }
  ],
  "intervention_trajectory": [
    {
      "step": "...",
      "timestamp": "...",
      "basis_weight": "...",
      "lower_bound": "...",
      "upper_bound": "...",
      "deviation_pct": "...",
      "lower_spec_limit": "...",
      "upper_spec_limit": "..."
    }
  ],
  "baseline_crossing_probability": 0.5,
  "intervention_crossing_probability": 0.5,
  "risk_reduction": 0.5,
  "expected_deviation_reduction": 0.5,
  "expected_stabilization_improvement": 1,
  "crossing_delay_steps": 1
}
```

GET /relationships/discovery

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Relationship Discovery

Parameter	Location	Required	Description
max_lag	query	no	integer
grade_pair	query	no	
stage	query	no	
method	query	no	
min_strength	query	no	number
limit	query	no	integer

Response example - HTTP 200

```
{}
```

Generate Recommendations

Request example

```
{
  "forecast_id": "00000000-0000-4000-8000-000000000001",
  "max_results": 5,
  "max_variables": 2
}
```

Response example - HTTP 201

```
[
  {
    "recommendation_id": "00000000-0000-4000-8000-000000000001",
    "forecast_id": "00000000-0000-4000-8000-000000000001",
    "state": "proposed",
    "rank": 1,
    "affected_variables": [
      "string"
    ],
    "current_values": {},
    "changes": [
      "...",
    ],
    "baseline_trajectory": [
      "...",
    ],
    "intervention_trajectory": [
      "...",
    ],
    "metrics": {
      "crossing_probability_before": "...",
      "crossing_probability_after": "...",
      "predicted_peak_deviation_before": "...",
      "predicted_peak_deviation_after": "...",
      "predicted_stabilization_time_before": "...",
      "predicted_stabilization_time_after": "...",
      "estimated_improvement": "...",
      "crossing_avoided": "...",
      "crossing_delay_steps": "..."
    },
    "confidence": 0.5,
    "constraint_validation": {
      "feasible": "...",
      "checks": "...",
      "violations": "...",
      "recipe_rules": "..."
    }
  }
]
```

Recommendation History

Parameter	Location	Required	Description
state	query	no	
limit	query	no	integer

Response example - HTTP 200

```
[
  {
    "recommendation_id": "00000000-0000-4000-8000-000000000001",
    "forecast_id": "00000000-0000-4000-8000-000000000001",
    "state": "proposed",
    "rank": 1,
    "affected_variables": [
      "string"
    ],
    "current_values": {},
    "changes": [
      "..."
    ],
    "baseline_trajectory": [
      "..."
    ],
    "intervention_trajectory": [
      "..."
    ],
    "metrics": {
      "crossing_probability_before": "...",
      "crossing_probability_after": "...",
      "predicted_peak_deviation_before": "...",
      "predicted_peak_deviation_after": "...",
      "predicted_stabilization_time_before": "...",
      "predicted_stabilization_time_after": "...",
      "estimated_improvement": "...",
      "crossing_avoided": "...",
      "crossing_delay_steps": "..."
    },
    "confidence": 0.5,
    "constraint_validation": {
      "feasible": "...",
      "checks": "...",
      "violations": "...",
      "recipe_rules": "..."
    }
  }
]
```

GET /interventions/recommendations/{recommendation_id} GradeSenseAI Honeywell Hackathon Submission

Get Recommendation

Parameter	Location	Required	Description
recommendation_id	path	yes	string

Response example - HTTP 200

```

{
  "recommendation_id": "00000000-0000-4000-8000-000000000001",
  "forecast_id": "00000000-0000-4000-8000-000000000001",
  "state": "proposed",
  "rank": 1,
  "affected_variables": [
    "string"
  ],
  "current_values": {},
  "changes": [
    {
      "variable": "...",
      "value": "..."
    }
  ],
  "baseline_trajectory": [
    {
      "step": "...",
      "timestamp": "...",
      "basis_weight": "...",
      "lower_bound": "...",
      "upper_bound": "...",
      "deviation_pct": "...",
      "lower_spec_limit": "...",
      "upper_spec_limit": "..."
    }
  ],
  "intervention_trajectory": [
    {
      "step": "...",
      "timestamp": "...",
      "basis_weight": "...",
      "lower_bound": "...",
      "upper_bound": "...",
      "deviation_pct": "...",
      "lower_spec_limit": "...",
      "upper_spec_limit": "..."
    }
  ],
  "metrics": {
    "crossing_probability_before": 0.5,
    "crossing_probability_after": 0.5,
    "predicted_peak_deviation_before": 0.5,
    "predicted_peak_deviation_after": 0.5,
    "predicted_stabilization_time_before": "...",
    "predicted_stabilization_time_after": "...",
    "estimated_improvement": 0.5,
    "crossing_avoided": true,
    "crossing_delay_steps": "..."
  },
  "confidence": 0.5,
  "constraint_validation": {
    "feasible": true,
    "checks": [
      "..."
    ],
    "violations": [
      "..."
    ],
    "recipe_rules": [
      "..."
    ]
  }
}

```

POST /interventions/recommendations/{recommendation_id}/decisions

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Decide

Parameter	Location	Required	Description
recommendation_id	path	yes	string

Request example

```
{
  "operator_action": "accepted",
  "reason": "string",
  "modified_values": {},
  "delay_duration_seconds": 1,
  "notes": "string"
}
```

Response example - HTTP 201

```
{
  "operator_action": "accepted",
  "reason": "string",
  "modified_values": {},
  "delay_duration_seconds": 1,
  "notes": "string",
  "decision_id": "00000000-0000-4000-8000-000000000001",
  "recommendation_id": "00000000-0000-4000-8000-000000000001",
  "timestamp": "2026-07-26T12:00:00Z",
  "state": "proposed"
}
```

POST /interventions/recommendations/{recommendation_id}/outcome GradeSense AI - Industrial paper grade transition intelligence

Evaluate Outcome

Parameter	Location	Required	Description
recommendation_id	path	yes	string

Request example

```
{
  "observations": [
    {
      "step": "...",
      "timestamp": "...",
      "basis_weight": "...",
      "lower_bound": "...",
      "upper_bound": "...",
      "deviation_pct": "...",
      "lower_spec_limit": "...",
      "upper_spec_limit": "..."
    }
  ]
}
```

Response example - HTTP 201

```
{
  "outcome_id": "00000000-0000-4000-8000-000000000001",
  "recommendation_id": "00000000-0000-4000-8000-000000000001",
  "metrics": {},
  "evaluated_at": "2026-07-26T12:00:00Z"
}
```

GET /interventions/effectiveness

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Effectiveness

Response example - HTTP 200

```
{
  "evaluated_count": 1,
  "crossing_avoidance_rate": 0.5,
  "crossing_delay_rate": 0.5,
  "mean_prediction_error": 0.5,
  "mean_deviation_improvement": 0.5,
  "mean_stabilization_improvement": 0.5
}
```

POST /demo/seed

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Seed Honeywell Demo

Response example - HTTP 201

```
{}
```

GET /admin/models

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Models

Response example - HTTP 200

```
[
{
  "version": "string",
  "name": "string",
  "model_kind": "prediction",
  "algorithm": "string",
  "trained_at": "2026-07-26T12:00:00Z",
  "dataset_checksum": "string",
  "feature_schema_checksum": "string",
  "artifact_checksum": "string",
  "artifact_path": "string",
  "metrics": {},
  "training_parameters": {},
  "description": "string"
}
]
```

GET /models

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Models

Response example - HTTP 200

```
[
{
  "version": "string",
  "name": "string",
  "model_kind": "prediction",
  "algorithm": "string",
  "trained_at": "2026-07-26T12:00:00Z",
  "dataset_checksum": "string",
  "feature_schema_checksum": "string",
  "artifact_checksum": "string",
  "artifact_path": "string",
  "metrics": {},
  "training_parameters": {},
  "description": "string"
}
]
```

GET /models/active

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Active Models

Response example - HTTP 200

```
[
{
  "version": "string",
  "name": "string",
  "model_kind": "prediction",
  "algorithm": "string",
  "trained_at": "2026-07-26T12:00:00Z",
  "dataset_checksum": "string",
  "feature_schema_checksum": "string",
  "artifact_checksum": "string",
  "artifact_path": "string",
  "metrics": {},
  "training_parameters": {},
  "description": "string"
}
]
```

GET /models/{model_id}

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Get Model

Parameter	Location	Required	Description
model_id	path	yes	string

Response example - HTTP 200

```
{
  "version": "string",
  "name": "string",
  "model_kind": "prediction",
  "algorithm": "string",
  "trained_at": "2026-07-26T12:00:00Z",
  "dataset_checksum": "string",
  "feature_schema_checksum": "string",
  "artifact_checksum": "string",
  "artifact_path": "string",
  "metrics": {},
  "training_parameters": {},
  "description": "string"
}
```

POST /models/register

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Register Model

Request example

```
{
  "version": "string",
  "name": "string",
  "model_kind": "prediction",
  "algorithm": "string",
  "trained_at": "2026-07-26T12:00:00Z",
  "dataset_checksum": "string",
  "feature_schema_checksum": "string",
  "artifact_checksum": "string",
  "artifact_path": "string",
  "metrics": {},
  "training_parameters": {},
  "description": "string"
}
```

Response example - HTTP 201

```
{
  "version": "string",
  "name": "string",
  "model_kind": "prediction",
  "algorithm": "string",
  "trained_at": "2026-07-26T12:00:00Z",
  "dataset_checksum": "string",
  "feature_schema_checksum": "string",
  "artifact_checksum": "string",
  "artifact_path": "string",
  "metrics": {},
  "training_parameters": {},
  "description": "string"
}
```

POST /models/promote

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Promote Model

Request example

```
{
  "model_id": "00000000-0000-4000-8000-000000000001"
}
```

Response example - HTTP 200

```
{
  "version": "string",
  "name": "string",
  "model_kind": "prediction",
  "algorithm": "string",
  "trained_at": "2026-07-26T12:00:00Z",
  "dataset_checksum": "string",
  "feature_schema_checksum": "string",
  "artifact_checksum": "string",
  "artifact_path": "string",
  "metrics": {},
  "training_parameters": {},
  "description": "string"
}
```

POST /models/archive

GradeSense | Honeywell Hackathon Submission

Archive Model

Request example

```
{
  "model_id": "00000000-0000-4000-8000-000000000001"
}
```

Response example - HTTP 200

```
{
  "version": "string",
  "name": "string",
  "model_kind": "prediction",
  "algorithm": "string",
  "trained_at": "2026-07-26T12:00:00Z",
  "dataset_checksum": "string",
  "feature_schema_checksum": "string",
  "artifact_checksum": "string",
  "artifact_path": "string",
  "metrics": {},
  "training_parameters": {},
  "description": "string"
}
```

GET /admin/metrics

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Metrics

Response example - HTTP 200

```
{}
```

GET /admin/config

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Config

Response example - HTTP 200

```
{
  "stream_speed_seconds": 0.5,
  "alert_thresholds": {},
  "forecast_horizon": 1,
  "history_window": 1,
  "confidence_threshold": 0.5,
  "feature_flags": {},
  "relationship_threshold": 0.5,
  "recommendation_limit": 1
}
```

PUT /admin/config

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Update Config

Request example

```
{
  "stream_speed_seconds": 0.5,
  "alert_thresholds": {},
  "forecast_horizon": 1,
  "history_window": 1,
  "confidence_threshold": 0.5,
  "feature_flags": {},
  "relationship_threshold": 0.5,
  "recommendation_limit": 1
}
```

Response example - HTTP 200

```
{
  "stream_speed_seconds": 0.5,
  "alert_thresholds": {},
  "forecast_horizon": 1,
  "history_window": 1,
  "confidence_threshold": 0.5,
  "feature_flags": {},
  "relationship_threshold": 0.5,
  "recommendation_limit": 1
}
```

GET /admin/health

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Admin Health

Response example - HTTP 200

```
{}
```

GET /admin/system

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Admin System

Response example - HTTP 200

```
{}
```

GET /admin/audit

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Audit

Parameter	Location	Required	Description
limit	query	no	integer

Response example - HTTP 200

```
[
  {
    "audit_id": "00000000-0000-4000-8000-000000000001",
    "timestamp": "2026-07-26T12:00:00Z",
    "actor": "string",
    "action": "string",
    "entity": "string",
    "entity_id": "string",
    "details": {},
    "request_id": "string"
  }
]
```

GET /admin/exports

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Exports

Response example - HTTP 200

```
[
  {
    "resource": "string",
    "formats": [
      "string"
    ],
    "row_count": 1
  }
]
```

POST /admin/export

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Create Export

Request example

```
{
  "resource": "forecasts",
  "format": "json"
}
```

Response example - HTTP 200

```
null
```

WebSocket Endpoints

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WS /ws/live and WS /ws/dashboard

Both sockets publish the same typed event envelope. The dashboard alias exists for compatibility.

```
{
  "event": "prediction",
  "timestamp": "2026-07-26T12:00:00Z",
  "data": {
    "quality_score": 0.94,
    "off_spec_risk": 0.08
  }
}
```

Event names

- system_status, heartbeat, sensor_update, prediction
- recommendation, alert, drift, basis_forecast
- recommendation_created, recommendation_updated, recommendation_decision, recommendation_outcome